

Statistical Inference

1. General

1.1 Title: Statistical Inference

1.2 Course Number (if known): MA455*

1.3 Credits: 3-1-0 (4 Credits)

1.4 Semester Offered: VII

1.5 Pre-requisites: MA-202: Probability and Statistics

1.6 Committee Members: Dr. Anuradha Banerjee and Dr. Amit Kumar

2. **Objectives:** The aim of this course is to apply probability theory and draw inferences about population parameters of interest.

3. Course Topics:

Unit I: Point Estimator (4 Lectures)

Definition of Estimator, Parameter Space, Unbiasedness, Consistency, Efficiency, Uniformly Minimum Variance Unbiased Estimator (UMVUE).

Unit II: Methods for Finding Estimators (6 Lectures)

Method of Moments, Method of Maximum Likelihood, Non-uniqueness of Maximum Likelihood Estimator (MLE), Method of Scoring for Finding Solution to Likelihood Equations, Invariance of MLE, Asymptotic Properties of MLE.

Unit III: Lower Bounds for Variance (6 Lectures)

Wolfowitz's Regularity Condition, Fréchet-Rao-Cramér Inequality, Efficiency of Estimator, Bhattacharya's Bound, Chapman-Robbins-Kiefer Inequality, Fréchet-Rao-Cramér Lower Bound in Higher Dimension.

Unit IV: Sufficiency and Completeness (5 Lectures)

Sufficiency, Rao-Blackwell Theorem, Neyman-Fisher Factorization Theorem, Application of Factorization Theorem, Minimal Sufficiency, Completeness, Boundedly Complete Statistic/Family, Rao-Blackwell-Lehmann-Scheffe Theorem, Ancillary Statistic, Basu's Theorem, Invariance Estimation Problem, Equivariant Estimator, Invariant Function, Maximal Invariant.

Unit VI: Bayes Estimator (4 Lectures)

Bayes Estimator, ε -Bayes Estimator, ε -minimax Estimator, Method for Finding Bayes Estimators, Limit of Bayes Estimator, Generalized Bayes Rule, Extended Bayes Rule, Relation Between Bayes and Minimax Estimator, Equalizer Estimator.

Unit VII: Testing of Hypothesis. (11 Lectures)

Null and Alternative Hypotheses, Tests of Statistical Hypotheses, Simple and Composite Hypotheses, Critical and Acceptance Region, Type I and Type II Errors, Power Function, Neyman-Pearson (NP) Lemma, Applications of NP Lemma, Family with monotone Likelihood Ratio, Most Powerful Test, Uniformly Most Powerful (UMP) Test, UMP

Invariant Test, UMP Unbiased Test for Exponential Family and Normal Population, Testing for Mean and Variance for Normal Population, Testing for Comparison of Means and Variances in Two Normal Population, Likelihood Ratio Test, Applications of Likelihood Ratio Test to One Sample and Two Samples Problem, Invariant Tests, UMP Invariant Test, Chi-square Test, Contingency Table, Test for Homogeneity, Sequential Probability Ratio Test.

Unit VIII: Interval Estimation

(3 Lectures)

Method for Finding Confidence Interval (CI), Examples for Several Population Including Normal Population, Expected Length of CI, Pivoting Method for Deriving CI, Shortest Length of CI, Unbiased CI, CI for Parameters of Two Normal Populations.

4. Textbooks and Reference Books

1. Rohatgi, V. K. and Saleh, A. K. Md. E., *An Introduction to Probability and Statistics*, 3rd Edition, John Wiley & Sons, 2015.
2. Casella, G. and Berger, R. L., *Statistical Inference*, 2nd ed. Belmont CA: Duxbury, 2002.
3. Kale, B. K., *A First Course on Parametric Inference*, Narosa Publishing House, 2002.
4. Mood, A.M., Graybill F. A. and Boes D. C., *Introduction to the Theory of Statistics*, McGraw Hill Education; 3rd edition, 2017.
5. Dudewicz, E. J. and Mishra, S. N., *Modern Mathematical Statistics*, Wiley Series in Probability and Statistics, 1988.

5. Other Sessions

- 5.1 Tutorials: Yes
5.2 Laboratory: No
5.3 Project: No

6. Assessment

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| 6.1 Quizzes | 10% |
| 6.2 Assignments | 10% |
| 6.3 Mid-Term Exam | 30% |
| 6.4 End-Term Exam | 50% |

7. Outcome of the course: After completion of this course, the students will be able to

- find estimators for the parameters of various populations and determine the best estimator among them.
- handle parametric hypothesis testing problems and apply the likelihood ratio method.
- perform interval estimation for the parameters of multiple populations.

8. Expected enrollment for the course: 70

9. Departments other than your own to which this course would be of interest: All Engineering Departments